

Stellarocq: Machine-Checked Certificates for Stellarator Equilibria

Charles C. Norton
cnorton2@binghamton.edu

Abstract

The output of a stellarator equilibrium code is a file of floating-point Fourier coefficients. Stellarocq states what such a file establishes as machine-checked theorems about the magnetic field it defines. The field is reconstructed from the exact numbers of a VMEC++ output file by VMEC's half-grid interpolation of the surface coefficients. The development proves that the residual it evaluates is $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$ of that field, and that its Jacobian and metric differ from those VMEC forms from the same numbers by exact terms of second order in the radial step. Each claim about the field, a bound, a Fourier harmonic, a flux-surface integral or a Mercier term, is decided by a checker extracted from a Rocq proof that a passing verdict implies the claim. The certificates give the error of the half-grid rule on manufactured problems to a relative 10^{-12} , falling by factors of 3.7 to 4.0 per halving of the radial step away from the axis. On VMEC++ output they bound away from zero, at 65 to 1025 surfaces, a harmonic of the force at a rational surface that falls at most 1.41-fold per doubling of the surfaces, approaching the square root of the radial step, where the axisymmetric harmonics fall about fourfold, and they enclose the free-boundary pressure jump for every vacuum field whose pressure lies at each point between the answers of NESTOR and BIEST. Against one of Landreman's exact three-dimensional equilibria they bound the difference between VMEC++'s field and the exact field at every point of grids on its flux surfaces, a difference that falls 3.5 to 5.6-fold per doubling of the surfaces away from the axis and stops falling at it, and the development proves in Rocq that the members of both his families are ideal-MHD equilibria. On W7-X they bound the force pointwise and at every retained mode, both set by the Fourier truncation, not the radial grid.

1 Introduction

The Variational Moments Equilibrium Code (VMEC) [1, 2] computes three-dimensional ideal-MHD equilibria with nested flux surfaces by minimizing the plasma energy over Fourier representations of the surfaces. It and its reimplementations VMEC++ [3, 4] sit at the centre of stellarator design and optimization. A run ends with a *wout* file: Fourier coefficients of the surfaces on a radial grid, profiles, and derived quantities such as the Mercier stability terms, all in binary64 floating point.

What such a file establishes about the field it describes is judged by convergence metrics internal to the solver and by diagnostics recomputed in floating point. Stellarocq states it as a theorem. The object of every statement is the field reconstructed from the exact numbers in the file by the solver's own interpolation of its surface coefficients, and the development proves how the Jacobian and metric of that field differ from the ones the solver forms. A claim about that field is decided by a checking function written in Rocq [5], proven to imply the claim when it returns `true`, and extracted to OCaml [6]; real-number semantics and interval arithmetic come from Coquelicot [7], Flocq [8] and CoqInterval [9].

The development contributes four things. It proves that the residual it evaluates is the ideal-MHD force of the reconstructed field and how that field's Jacobian and metric differ from the ones VMEC forms (Section 2). It supplies checking functions, extracted from Rocq together with their

soundness theorems, for bounds at points and over cells, discrete Fourier harmonics, flux-surface integrals, the Mercier criterion and the existence of discrete solutions (Section 3). It applies them to VMEC++ (Section 4): the error of the half-grid rule on manufactured problems, known to a relative 10^{-12} ; a harmonic of the force on a rational surface, bounded away from zero over a box of states; the free-boundary pressure jump, enclosed for every vacuum field between the answers of two independent solvers; the Mercier criterion of a coarse equilibrium, whose sign depends on how the radial derivatives are taken; the distance of VMEC++'s field from an exact three-dimensional equilibrium; and the force residual of W7-X, which its Fourier truncation sets. Its certified values are regression tests on a branch of VMEC++ [28]; its differential testing added tests of invariances and of the energy's gradient to VMEC++, and VMEC++'s continuous integration now measures its solutions against the exact equilibria [33].

2 The certified object

Reconstruction. VMEC keeps the surface coefficients of R and Z on the full radial grid $s_j = jh$ and the stream function λ , the rotational transform ι and every field quantity on the half grid $s_{j+1/2}$. Stellarocq's `Physics.v` writes the rule that connects them. On a half point s_h between nodes a and b , a coefficient of even poloidal mode number is the mean of its two node values and one of odd mode number is averaged with the $\sqrt{s}$ weight VMEC folds into its Jacobian,

$$c_k(s_h) = \frac{1}{2}(c_k(a) + c_k(b)) \quad (m_k \text{ even}), \quad c_k(s_h) = \frac{1}{2}\sqrt{s_h} \left(\frac{c_k(a)}{\sqrt{s_a}} + \frac{c_k(b)}{\sqrt{s_b}} \right) \quad (m_k \text{ odd}), \quad (1)$$

with radial derivatives the matching difference quotients and angular derivatives exact derivatives of the series $R = \sum_k c_k \cos(m_k u - n_k v)$, $Z = \sum_k z_k \sin(m_k u - n_k v)$. The half-grid λ and ι are the file's own, Φ' is constant, and the pressure is the input's profile. For statements at an arbitrary radius each coefficient of R and Z is the cubic Hermite interpolant of its half-point values and slopes, which agrees with the half-grid rule at the half points; λ follows the same parity rule linearly between half points, and ι is linear there.

Field and residual. With $\sqrt{g} = R(R_u Z_s - R_s Z_u)$, the field is $B^u = \Phi'(\iota - \lambda_v)/\sqrt{g}$, $B^v = \Phi'(1 + \lambda_u)/\sqrt{g}$, its covariant components are $B_i = g_{iu} B^u + g_{iv} B^v$, and $\mu_0 \sqrt{g} J^s = \partial_u B_v - \partial_v B_u$. The residual of $\mathbf{J} \times \mathbf{B} = \nabla p$ has the components

$$r_u = -\mu_0 \sqrt{g} J^s B^v, \quad r_v = \mu_0 \sqrt{g} J^s B^u, \quad r_s = (\partial_v B_s - \partial_s B_v) B^v - (\partial_s B_u - \partial_u B_s) B^u - \mu_0 \frac{dp}{ds}, \quad (2)$$

written in `Physics.v` as trees of a deep expression language. Read at a free radius, with every radial derivative that of the Hermite reconstruction, they form the continuum residual. Let $X(s, u, v) = (R \cos v, R \sin v, Z)$ be the reconstructed coordinate map, $\mathbf{e}_s$, $\mathbf{e}_u$ and $\mathbf{e}_v$ its partial derivatives, and $\mathbf{B} = B^u \mathbf{e}_u + B^v \mathbf{e}_v$ the reconstructed field as a function of (s, u, v) .

Theorem 1 (continuum_force_coords) *Let the slots of the residual's expressions hold a state's radii, coefficients, flux, transform and pressure derivative p' , and let (s_0, u_0, v_0) be a point where $\sqrt{g} \neq 0$. There let $\nabla s = \mathbf{e}_u \times \mathbf{e}_v / \sqrt{g}$, $\nabla u = \mathbf{e}_v \times \mathbf{e}_s / \sqrt{g}$ and $\nabla v = \mathbf{e}_s \times \mathbf{e}_u / \sqrt{g}$, and let $\partial_s \mathbf{B}$, $\partial_u \mathbf{B}$ and $\partial_v \mathbf{B}$ be the derivatives of $\mathbf{B}$ along the three coordinate lines through the point. Then the three outputs of the continuum residual there are $\mathbf{F} \cdot \mathbf{e}_s$, $\mathbf{F} \cdot \mathbf{e}_u$ and $\mathbf{F} \cdot \mathbf{e}_v$, with*

$$\mathbf{F} = (\nabla s \times \partial_s \mathbf{B} + \nabla u \times \partial_u \mathbf{B} + \nabla v \times \partial_v \mathbf{B}) \times \mathbf{B} - \mu_0 p' \nabla s. \quad (3)$$

The bracket is the curl of $\mathbf{B}$ in the coordinates $x^j = s, u, v$, since the Cartesian derivative of $\mathbf{B}$ is $\sum_j \nabla x^j \otimes \partial_j \mathbf{B}$. Every quantity in the statement is a formula in the reconstruction's derivatives at the point, and with $\nabla p = p' \nabla s$ the outputs are the covariant components of $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$. `continuum_force` gives the same outputs through the Cartesian curl: for any $\mathbf{G} : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ and $P : \mathbf{R}^3 \rightarrow \mathbf{R}$ differentiable at $X(s_0, u_0, v_0)$ and equal to the reconstructed field and pressure along the three coordinate lines through it, they are $\mathbf{F} \cdot \mathbf{e}_i$ with $\mathbf{F} = (\nabla \times \mathbf{G}) \times \mathbf{G} - \mu_0 \nabla P$. `continuum_force_asym_coords` and `continuum_force_asym` are the two statements for a state without stellarator symmetry, and they rest on the identities at one point `Force.force_coords` and `Force.force_law`. At a node the solver's rule is used instead: the angular outputs are the formulas above at the outer half point, the radial output is VMEC's combination of the fields at the two half points, with centred differences for radial derivatives, and each of those fields is that of the series with VMEC's half-grid values and slopes (`node_residual`, `node_residual_asym`).

VMEC's products. Every half-point quantity that is linear in the coefficients, R , Z and their derivatives, agrees with VMEC's term for term. The Jacobian and the metric elements are products, which VMEC forms from averages over the two nodes of products of node values, where the reconstruction multiplies the interpolated series. `VmecKernel.v` transcribes VMEC++'s kernels and proves the differences exactly. With r, z the even parts and $\hat{r}, \hat{z}$ the odd parts divided by $\sqrt{s}$ of the node values, and Δ the outer node's value less the inner one's, the reconstruction's factor $R_u Z_s - R_s Z_u$ and VMEC's τ satisfy

$$(R_u Z_s - R_s Z_u) - \tau = -\frac{1}{8}(\Delta \hat{r}_u \Delta \hat{z} - \Delta \hat{z}_u \Delta \hat{r}) - \frac{1}{8\sqrt{s_h}}(\Delta r_u \Delta \hat{z} - \Delta z_u \Delta \hat{r}) \quad (4)$$

(`vmec_jacobian_difference`), and each of g_{uu} , g_{uv} and g_{vv} differs from VMEC's by a sum of such products (`vmec_metric_difference`). Where every node difference in the identity is at most $K|h|$, the two Jacobian factors agree to within $\frac{1}{4}(Kh)^2(1+s_h^{-1/2})$ (`vmec_jacobian_second_order`). The certified field is the reconstruction's, and VMEC's discrete field differs from it by these terms of second order in the radial step. Evaluated in floating point on the files of Section 4 (`gen/jacobian_gap.py`), the two Jacobians, and so the two $\sqrt{g}$ (`vmec_sqrtg_difference`), differ by at most a relative 6.3×10^{-7} , 1.7×10^{-7} , 4.2×10^{-8} , 4.5×10^{-9} and 1.2×10^{-9} on the surfaces with $s \geq 1/4$ of Landreman's sheared equilibrium at 13, 25, 50, 100 and 200 surfaces, and by at most 1.1×10^{-6} on every half-grid surface whose nodes lie off the axis; on the Solov'ev equilibrium the largest is 9.1×10^{-7} .

Identities. The representation builds some properties in, and for the reconstruction they are theorems: it is solenoidal (`divergence_free`), the $(0, 0)$ coefficient of λ drives nothing (`lambda_gauge`), which exhibits the poloidal relabelling as an exact kernel direction, and the residual has no component along the field (`residual_along_field`). The Euler-Lagrange expressions of VMEC's energy $W = \int (B^2/2 - \mu_0 p) \sqrt{g}$ at fixed λ , ι , flux and pressure are $-\sqrt{g}$ times the cylindrical components of the force (`energy_gradient_is_force`), so that the forces the solver drives to zero are the ideal-MHD force. `Hypotheses.v` states the physical premises some theorems carry, such as nested flux surfaces and a scalar pressure, and the second-order bound `discretization_is_consistent` names, which `HalfGrid.v` derives from the proven consistency of the node rule and a stability bound (Section 4).

3 Certificates and their soundness

A certificate is a text file that a generator in `gen/` writes from a wout file. It carries every number of the file it uses as an integer mantissa and a power-of-two exponent, so no rounding happens between the file and the certificate, together with the bounds it claims; the extracted checker recomputes every enclosure from those numbers, and no bound the generator computes is trusted.

Expressions and intervals. `Expr.v` gives an expression two meanings: a value over extended reals, in which division by zero and the square root of a negative number give `Xnan`, and a value over CoqInterval’s floating-point intervals. `ieval_correct` states that the interval contains the extended-real value, so a bounded output interval proves both that the value is a real number and that it lies in the bounds. Because the embedding is deep, one tree carries both meanings, `Deriv.v` differentiates it symbolically with a chain rule proven in the extended-real sense, and subexpressions that the residual’s components share are bound once to environment slots.

Verdicts. A point certificate bounds the residual at points (`check_cert_correct`). A cell certificate bounds a component over a rectangle of two varied slots, two angles or an angle and the radius, by its value at the centre plus enclosures of its derivatives over the rectangle times the half-widths (`check_ccert_correct`); reversed, the same data bounds a component away from zero. The bound tightens as the cells shrink, since each derivative enclosure is taken over its own cell: on one surface of the Solov’ev equilibrium, where the residual reaches 4.1×10^{-7} in floating point, 8, 32, 128, 512, 2048 and 8192 cells bound it by 2.2×10^3 , 0.97, 3.4×10^{-2} , 2.0×10^{-3} , 1.3×10^{-4} and 7.7×10^{-6} , about sixteen-fold less per fourfold refinement from 128 cells on, in 0.4 to 13 seconds. `Cover2.v` decides, column by column, that the cells of a verdict cover the rectangle they span (`cells_cover_rect`) or every line they lie on (`cells_cover_lines`). A spectral solution balances forces mode by mode, so the quantity it makes small is the discrete Fourier harmonic of the residual at a retained mode, which `Project.v` encloses (`harm_encloses`); `Harmonic.v` integrates exactly the components whose trigonometric degree lies below an equispaced grid’s point counts, and `Integrate.v` encloses other flux-surface integrals by the midpoint rule over a lattice of cells that the checker lays out itself, with the error read off its own enclosures of the second derivatives over each cell (`integ2_correct`, `integ1_correct`). `MercierRun.v` assembles DShear, DCurr, DWell, DGeod and DMerc as VMEC’s `mercier.f90` defines them [10, 11] from integrals over the whole angular torus that the same run computes: the lattice reaches past each period of 2π , the period ends inside its last cells, and the two strips it adds are bounded through the integrand’s enclosures over those cells (`integ2_torus_correct`). A verdict is then a theorem about the criterion of the reconstruction (`merc_run_correct`). In `Box.v` every input slot carries a half-width, so one evaluation covers every field whose coefficients lie in a box. The theorem behind the resonant harmonic of Section 4 shows the form every soundness theorem takes:

Theorem `dharm_correct` :

```
forall c, check_dharm c = true ->
forall xs, in_box (hc_ms c) (hc_ds c) xs ->
contains (I.convert (harm_total c))
  (Xreal (dsum2 (hc_Nu c) (hc_Nv c) 0 0 (harm_fun c xs))).
```

For a certificate `c` whose check passes and every state `xs` in its box, the equispaced sum of the component times the harmonic’s kernel over the certificate’s grid of angles is a real number, and it lies in the interval `harm_total c` that the same computation returns.

Discrete equilibria. `Newton.v` proves that a passing interval Newton test in Krawczyk form [12, 13] on any system of expressions gives exactly one zero in its box: $x - AF(x)$ maps the box into itself as a contraction, and an approximate inverse A makes its fixed point a zero of F . `Colloc.v` collocates the residual over a band of surfaces as such a system, with the surface coefficients as unknowns (`colloc_correct`), and `Wide.v` evaluates the system’s outputs at the centre of the box in floating point of any precision.

Trust base. The kernel checks the proofs; what runs is the OCaml extraction of the checking functions, linked with a driver that parses certificates into the extracted data types, divides the checking among processes and prints verdicts. The trusted components are the Rocq kernel and extraction, the OCaml toolchain, the implementation of Rocq’s primitive 63-bit integers and binary64 floats, the driver’s parser and its division of the work, and the correspondence between a certificate’s numbers and the output or formulas they are taken from, which `gen/verify_cert.py` checks for a wout file. The classical constructions behind Rocq’s real numbers appear only in specifications, and extraction directives replace them by values that fail when consulted. The theorems named here rest on the two axioms of the standard library’s classical Dedekind reals, excluded middle, functional extensionality and the specifications of the primitive operations. The build pins Rocq 9.1.1 with its standard library 9.2.0, CoqInterval 4.11.4, Flocq 4.2.2, Coquelicot 3.4.5, MathComp 2.4.0 and OCaml 4.14.2.

4 Results

Table 1 collects the certified results with the theorem each rests on. The manufactured problems are Stellarocq’s collocated systems for the manufactured mappings of the VMEC++ pull request [29] that adds a force source to the solver, and every other case is a VMEC++ run or a file from its test data. The runs of Landreman’s equilibrium, of the rippled tokamak and of W7-X are VMEC++ 0.4.7, and the free-boundary and Mercier runs the branch of [28], which reads NESTOR’s vacuum field. A number called certified is an enclosure that a passing verdict proves, rounded outward.

The order of the half-grid rule. One, two and three surfaces of the Solov’ev equilibrium [14] each have exactly one zero of the collocated residual in a box shaped to the coefficients, of radius 6.0×10^{-10} to 7.7×10^{-9} , around a centre that Newton’s method moved 1.8×10^{-6} to 3.0×10^{-6} from the file’s coefficients. `HalfGrid.v` proves the node rule second-order consistent with the residual of the continuously reconstructed field (`node_consistent`) and derives the second-order bound that `discretization_is_consistent` states from consistency and a stability hypothesis (`lax_second_order`); near the axis, where the innermost interval is read linearly, the rule is first order in the radial derivatives. Manufactured solutions [15] measure the order. The mappings are the $m \leq 1$ ones of the manufactured-solution test of [29], with three field periods. The three-dimensional one is $R = 1 + p_0 s + p_1 s^2 - 0.005 \cos 3\phi + \sqrt{s} (0.05 \cos u + 0.005 \cos(u - 3\phi))$ and $Z = 0.005 \sin 3\phi + \sqrt{s} (0.05 \sin u - 0.005 \sin(u - 3\phi))$ with $p_0 = -3.32 \times 10^{-4}$ and $p_1 = -8.06 \times 10^{-6}$, $\iota = 0.42 + 0.13s$, $p = 1.6 \times 10^5 (1 - s)$ Pa and a stream function with harmonics up to $m = 2$, and the one without stellarator symmetry adds $0.004 \sin 3\phi + \sqrt{s} (0.006 \sin u + 0.002 \sin u \cos 3\phi)$ to R and $0.003 \cos 3\phi + \sqrt{s} (0.006 \cos u + 0.002 \cos u \cos 3\phi)$ to Z . The file that defines them, `examples/manufactured_solution.py` of that pull request at commit `fdbb696`, is copied into Stellarocq’s `gen/`, and the generator reads every constant of the problem from it. A manufactured mapping on the annulus $1/4 < s < 1$ is not an equilibrium; its continuum residual at the collocation points is a source f , and the zero x_h of $r - f$ over the interior nodes, with λ given so that the

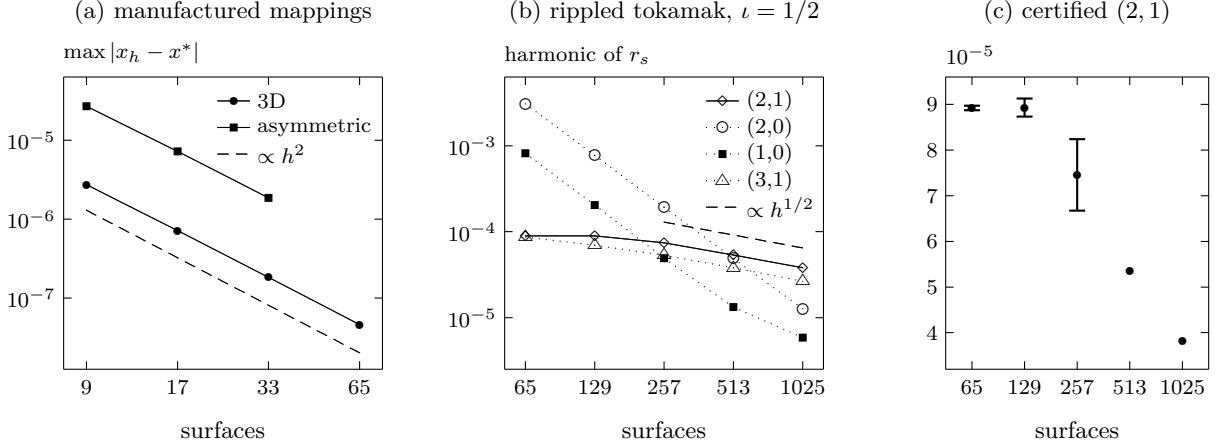


Figure 1: (a) The certified error of the half-grid scheme on the manufactured mappings; the dashed line falls as h^2 . (b) Harmonics of the radial force r_s on the surface $\iota = 1/2$ of the rippled tokamak: the resonant (2,1) harmonic, certified at 256 bits, and the floating-point (2,0), (1,0) and (3,1) harmonics of the same reconstruction; the dashed line falls as $h^{1/2}$. (c) The (2,1) harmonic on a linear scale: its binary64 enclosures at 65, 129 and 257 surfaces, and its 256-bit enclosures, narrower than the dots, at 65 to 1025.

problem sits on the quotient by the poloidal gauge, is the discrete solution. Every datum of the problem, the coefficients it holds fixed, λ , the transform and the source, is the mapping's own value carried to 120 bits. On systems of up to 423 unknowns the interval Newton test puts exactly one zero in a box a few units of a 58-bit mantissa wide in each unknown, around a centre refined on outputs read at 128 bits, so the error $\max |x_h - x^*|$ is enclosed to within 5×10^{-20} , a relative 10^{-12} ; the same problem with its data rounded to binary64 has an error that differs in the eighth digit. It falls by 3.81, 3.92 and 3.97 per doubling for a three-dimensional mapping and by 3.73 and 3.87 for one without stellarator symmetry (Figure 1a), and its enclosures are identical at five times the pressure, which enters the radial force only as a flux function that the rule and its source share. The collocated system is the half-grid residual of the reconstruction at collocation points. VMEC's own discrete equations come from the variation of the energy with spectral condensation and a constraint on the $m = 1$ modes, and they form the Jacobian and metric from the products whose difference from the reconstruction's `VmecKernel.v` gives exactly, at second order in h . The annulus keeps the axis, where the rule is first order, out of the problem. The first-order radial convergence that [16] report for VMEC is that of the force error of VMEC's own solutions averaged over $0.1 \leq s \leq 0.99$ of a discrete problem that contains the axis.

A forced current sheet. On VMEC++'s circular tokamak ($R_0 = 6$, $a = 2$) with $\iota = 0.9 - 0.64s$, which is $1/2$ at $s = 0.625$, and a (1,1) boundary ripple of amplitude 0.03, toroidal coupling drives a (2,1) harmonic of the radial force on the rational surface. Its certified value is the rectangle rule for the integral of $r_s \cos(2u - v)$ on a 64×32 grid of angles, in T^2 like r_s , against $1.2 \times 10^3 T^2$ for the integral of B^2 over the angles there. With each row of the grid evaluated and summed at 256 bits, `wdharm_b_correct` encloses it at the converged states of 65, 129, 257, 513 and 1025 surfaces to within a relative 2×10^{-12} of 8.9208×10^{-5} , 8.9310×10^{-5} , 7.4567×10^{-5} , 5.3557×10^{-5} and 3.8084×10^{-5} (Figure 1c), and it is linear in the ripple. It rises by 0.11 per cent from 65 to 129 surfaces and then falls by 1.20, 1.39 and 1.41 per doubling, approaching the factor $\sqrt{2}$ of a decay as

the square root of the radial step, while the axisymmetric harmonics $(2, 0)$ and $(1, 0)$ of the same reconstruction fall 3.7 to 4.1-fold per doubling in floating point up to 513 surfaces, and 3.8 and 2.3-fold from 513 to 1025, and the $(3, 1)$ sideband 1.24 to 1.44-fold per doubling (Figure 1b). In floating point the harmonic forms a layer about the rational surface, positive inside it and negative outside, whose extremes stay of order 10^{-4} T^2 while their separation shrinks only as $h^{1/2}$, from 0.156 to 0.033 in s between 65 and 1025 surfaces, and whose zero lies a fixed 2.4 to 3.9 radial steps outside the rational surface, so that the value on that surface, the layer's slope times the offset, falls as $h^{1/2}$. The same ripple on a transform profile with no low-order rational surface nearby leaves nothing behind. This is the response of a nested-surface field at a rational surface [17, 18, 19], whose ideal limit is a current sheet and which [20] studied in VMEC; three-dimensional equilibria with current sheets and stepped pressure exist [21, 22]. In binary64 interval arithmetic (`dharm_correct`) the enclosures at 65, 129 and 257 surfaces widen about fourfold per doubling, from 1.1 to 21 per cent of the value, as second radial differences amplify the rounding by h^{-2} , and there the harmonic lies between 5.8×10^{-5} and 9.4×10^{-5} for every state within a relative 10^{-14} of the converged coefficients. The box is centred on the converged state, and on the rational surface the collocated Newton test, its centre refined on outputs read at 128 bits, certifies a discrete solution at 33 and at 65 surfaces, where the Jacobian's condition number is 1.4×10^6 and 3.0×10^6 in floating point. In SPEC [23], two Beltrami volumes separated by an ideal interface at the rational surface carry the $(2, 1)$ sheet as a jump of 1.47×10^{-5} in the covariant poloidal field at ripple 0.03, as in the equilibria of [24], and with the rational surface inside a relaxed volume SPEC's field-line tracing finds a $(2, 1)$ island whose width grows from 0.107 to 0.173 in the volume's radial coordinate as the ripple goes from 0.0075 to 0.03, as the ripple to the power 0.35.

Free-boundary balance. At a free-boundary equilibrium $\mu_0 p + B^2/2$ is continuous across the plasma boundary, where outside it is the pressure of the full vacuum field. VMEC++ computes that field with NESTOR [25, 2], and BIEST's virtual-casing field [26, 27] gives it independently. The certificate sweeps the vacuum pressure over the segment between the two answers, $P_{\text{vac}} = P_{\text{mid}} + t \delta P$ with $t \in [-1, 1]$, and takes the plasma side as VMEC's extrapolation of the two outermost half points to $s = 1$. The sweep variable is a slot of the box, which `check_bpcert_correct` and `check_btcert_correct` quantify over separately at each point and in each cell, so t may differ from point to point: every statement below holds for each vacuum pressure whose value at each point lies between the two answers there. On the CTH-like test case the largest jump on NESTOR's grid is enclosed in $[7.556, 7.561]$, $[7.201, 7.206]$, $[6.433, 6.437]$, $[5.636, 5.640]$ and $[4.9647, 4.9673] \times 10^{-4} \text{ T}^2$ at 25, 49, 97, 193 and 385 surfaces, for every t , where $B^2/2$ averages 0.18 T^2 on the boundary. Each enclosure lies below the one before it, so the jump falls under radial refinement, but only by 1.05 to 1.14 per doubling. The two answers differ by up to $6.1 \times 10^{-4} \text{ T}^2$ on the grid, and with the sweep widened to five times that difference, $P_{\text{vac}} = P_{\text{mid}} + 5t \delta P$, the largest jump is at most 1.87, 1.89, 1.81, 1.73 and $1.66 \times 10^{-3} \text{ T}^2$ at 25 to 385 surfaces, a hundredth of $B^2/2$, for every vacuum pressure within that band at each point. VMEC imposes the balance through the Fourier harmonics it retains, and in floating point the jump on NESTOR's grid lies almost wholly beyond them: at 49 surfaces its part in the retained harmonics is at most 2.4×10^{-5} and the rest 7.2×10^{-4} . The retained part falls 3.3 and 1.9-fold per doubling from 25 to 97 surfaces, while with NESTOR's poloidal grid fixed, retaining 5 to 12 poloidal harmonics moves the jump between 4.0×10^{-4} and 1.5×10^{-3} without reducing it. Covered over the whole boundary by 512×128 cells per field period, with the vacuum side the trigonometric interpolant of each answer's grid, the jump is at most 8.47×10^{-3} at 25 surfaces and 1.051×10^{-2} at 385, 11 and 21 times the bound on the grid. Between grid points the bound is set by that interpolant, whose angular resolution radial

refinement leaves as it is, so it does not fall.

The Mercier criterion at coarse resolution. At $s = 0.125$ of `up_down_asym` on 17 surfaces, with 2048 poloidal cells, the certified Mercier current term lies in $[-3.724, -3.716] \times 10^{-3}$ where the file's difference quotients give -6.1×10^{-4} , and the certified DMerc lies in $[-1.371, -1.205] \times 10^{-3}$ where the file's is $+1.8 \times 10^{-3}$: the reconstruction's criterion is certified unstable where the file's is stable. From 33 to 257 surfaces both current terms tend to zero with the file's sign, the certified one from $[-1.092, -1.083] \times 10^{-3}$ to $[-3.7, -1.8] \times 10^{-5}$, and the certified DMerc is positive at each. At 17 surfaces the radial derivatives set the sign, and both signs are certified. `MercierFile.v` takes the four flux-function derivatives from the file's centred differences of its half-grid profiles, each an exact dyadic, and integrates the four angular integrands as before (`merc_run_q_correct`): the current term then lies in $[-6.421, -6.366] \times 10^{-4}$ and DMerc in $[1.711, 1.875] \times 10^{-3}$, which contains the file's floating-point value, where the Hermite reconstruction's derivatives give the certified negative DMerc above.

An exact equilibrium. Landreman's families [58] are three-dimensional equilibria with nested flux surfaces whose field $\mathbf{B}^*$ and flux label ψ^* are elementary functions of the Cartesian position. `Landreman.v` and `LandremanSheared.v` prove that both consist of ideal-MHD equilibria: the family with $\iota = 2$ for all positive semiaxes wherever the radicand of $\mathbf{B}^*$ is positive, and the sheared family for all positive ϵ and λ and every S wherever (x, y) lies off the segment $x = 0, |y| \leq \sqrt{\epsilon}$. In each the field is solenoidal (`iota2_divergence`, `sheared_divergence`), satisfies $(\nabla \times \mathbf{B}) \times \mathbf{B} = \nabla p$ with $p = p_a - 2\psi$ and $p = p_a - \psi/\lambda^2$ respectively (`iota2_force`, `sheared_force`), and is tangent to the level sets of p (`iota2_tangent`, `sheared_tangent`). Two outputs of the checker compare the reconstruction with a member of either family: the cylindrical components of $\mathbf{B} - \mathbf{B}^*(X)$ on a half-grid surface, with $\mathbf{B}$ the reconstructed field and X its position, and $\psi^*(X)$ on a node surface. `LandremanExact.v` proves that the expressions behind those outputs evaluate to the field and flux label of `Landreman.v` and of `LandremanSheared.v` on those domains (`exact_b_iota2`, `exact_b_sheared`). VMEC++'s solution of the sheared member with $\epsilon = 0.6$, $S = 2.2$, $\lambda = 2.97$ and $k_b = 0.2$, whose transform runs from 4.34 to 4.37, at 13 surfaces with $m \leq 11$ and $|n| \leq 10$, and at 25, 50, 100 and 200 surfaces, is certified at every point of a 64×32 grid of a field period on its innermost, middle and outermost half-grid surfaces, where $|\mathbf{B}^*|$ lies between 0.30 and 0.38 T. On the outermost the largest component of $\mathbf{B} - \mathbf{B}^*$ is at most 8.25×10^{-6} , 2.00×10^{-6} , 5.67×10^{-7} , 1.02×10^{-7} and 2.12×10^{-8} T at 13, 25, 50, 100 and 200 surfaces, and on the middle one, at s near $1/2$, 1.32×10^{-5} , 2.93×10^{-6} , 5.39×10^{-7} , 1.12×10^{-7} and 2.53×10^{-8} T, falling 3.5 to 5.6-fold per doubling; on the innermost, at $s = 0.125, 0.062, 0.031, 0.015$ and 0.008 , it is at most 2.94×10^{-5} , 1.36×10^{-5} , 5.61×10^{-6} , 4.35×10^{-6} and 3.34×10^{-6} T, and stops falling at the axis. The outermost node surfaces lie within 2.32×10^{-7} , 4.95×10^{-8} , 1.77×10^{-8} , 1.70×10^{-9} and 1.71×10^{-10} of a level set of ψ^* , whose value at the boundary is 0.02. Against the member with $\iota = 2$, whose every surface is rational, VMEC++'s field does not converge: in floating point its distance from $\mathbf{B}^*$ at mid-radius is 8.4×10^{-4} , 2.3×10^{-4} , 1.9×10^{-3} and 1.9×10^{-4} T at 13, 25, 50 and 100 surfaces, with 77 to 86 per cent of its spectral power in the resonant harmonics, which bend no field line when $\iota = 2$ on every surface, so that VMEC's energy is flat along them.

An optimized stellarator. VMEC++'s solution of W7-X from its test data, at a volume-averaged β of 4.0 per cent with $m \leq 11$ and toroidal numbers up to 12 per field period, converged to $f_{sqr} < 10^{-12}$ at 49, 97 and 193 surfaces, is certified at every point of a 64×64 grid of a field period on the surfaces $s = 1/4, 1/2$ and $3/4$ (`check_cert_correct`), and the same verdicts enclose the

discrete harmonics of each component there at the 288 modes VMEC++ retains (`harm_encloses`). The largest component of $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$ is at most $7.71, 7.73$ and $7.82 \times 10^{-2} \text{ T}^2$ at 49, 97 and 193 surfaces, against a B^2 of about 7 T^2 , and the largest harmonic at a retained mode at most $5.49, 5.49$ and $5.46 \times 10^{-4} \text{ T}^2$, at $m = 10$, next to the edge of the retained band; neither falls as the surfaces double. With $m \leq 15$ and $m \leq 19$ at 49 surfaces, toroidal numbers up to 16 and 20, the largest component is at most 5.72 and $3.84 \times 10^{-2} \text{ T}^2$ and the largest retained harmonic at most 3.66 and $1.51 \times 10^{-4} \text{ T}^2$. The residual of this solution is set by its Fourier truncation and not by its radial grid.

Use in VMEC++. The certified values are regression tests on a branch of VMEC++ [28], whose tolerances are the enclosures: the manufactured-solution errors, the resonant harmonic with the fall of the axisymmetric harmonics, and the free-boundary jump, which reads NESTOR’s boundary vacuum field through an option proposed upstream [30]. A finite difference of VMEC++’s energy equals its unpreconditioned forces contracted with the perturbation, the discrete form of `energy_gradient_is_force`; VMEC++ now tests this [31], and that the energy Hessian is symmetric and DGeod never positive [32]. Its continuous integration now runs VMEC++ on members of both of Landreman’s families on every change and measures, in floating point, its flux surfaces, magnetic axis, field and enclosed current against the exact ones [33]. Differential tests of VMEC++ for invariances the reconstruction satisfies exactly, three of which VMEC++ now runs [34], also exposed an orientation test that the poloidal rotation VMEC applies to an asymmetric boundary can defeat: the test multiplies two $m = 1$ coefficients whose product the rotation can drive to zero, where the invariant is the $m = 1$ determinant $r^{cc}z^{sc} - r^{sc}z^{cc}$. Relabelling the `up_down_asym_current` boundary by $\theta \rightarrow \pi - \theta$ sends educational_VMEC to its iteration cap at $f_{sqr} = 2.07 \times 10^{-2}$ instead of converging to 5.07×10^{-12} . It was reported as vmecpp#788 [35], after which VMEC++ tests the determinant [36], and as educational_VMEC#27 [37].

5 Related work

Rigorous numerics proves statements about dynamical systems and partial differential equations with interval arithmetic: the existence of the Lorenz attractor [38], the CAPD library for ordinary differential equations [39], KAM theory applied by computer [40], among other systems to magnetic field lines [41], blow-up of the Boussinesq and axisymmetric Euler equations [42], and elliptic and parabolic problems [43], among them the existence of stationary Navier–Stokes flows in three-dimensional domains near a computed approximation [44]. These proofs run in libraries outside a proof assistant. Inside one, verified ODE solvers up to the Lorenz attractor [45, 46], the discretization and floating-point program of the one-dimensional wave equation [47], the Lax–Milgram theorem [48] and simplicial Lagrange finite elements [49] in Rocq, and the certificates of the Kepler conjecture [50] come closest; in Lean, high-energy physics is being formalized [51] and a formalization has found an error in the literature on the stability of the two-Higgs-doublet potential [52], the characterization of the equilibria of the Vlasov–Maxwell–Landau system has been formalized [53], and smooth magnetohydrostatic equilibria with nested pressure surfaces and only a discrete rotational symmetry, counterexamples to Grad’s conjecture [17], come with a Lean certification [54]. In fusion, formal methods have verified a plasma vertical-position control algorithm [55] and properties of neural surrogates of the Grad–Shafranov equation [56]. Stellarator equilibrium codes are verified by convergence studies, manufactured solutions and comparison between codes, VMEC against SPEC [23, 24, 20] and against DESC [57, 16], and against Landreman’s explicit three-dimensional equilibria [58], whose supplement solves them in DESC. We know of no prior

work that states the output of a stellarator equilibrium code as a machine-checked theorem about the field it defines.

6 Conclusion

Stellarocq gives the output of an equilibrium code a meaning a proof assistant can check: bounds, harmonics, stability terms and the existence of discrete solutions, each a theorem about the field reconstructed from the file's exact numbers, whose residual is proven to be the ideal-MHD force. Its verdicts are regression tests on a branch of VMEC++ whose tolerances are the certified enclosures. Against Landreman's explicit equilibria [58], both of whose families the development proves to be ideal-MHD equilibria, the same certificates measure a solver's error against a true equilibrium rather than a manufactured one, a comparison VMEC++ now makes in floating point on every change [33].

Availability. The development, the certificate generators and the script that writes Figure 1 are at <https://github.com/CharlesCNorton/stellarocq>, commit 18b64aa; `make all` builds the proofs and the checker, the release `checker-18b64aa` carries the same checker linked statically for x86_64 Linux with the certificates behind the results and their SHA-256 sums, and each generator's documentation gives the commands of its result. The VMEC++ tests that carry the certified values are on the branch of [28].

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Case	Certified quantity	Enclosure
manufactured, 3D	$\max x_h - x^* $, 9/17/33/65 surfaces colloc_correct	$2.73432936829 \times 10^{-6}$, $7.16912170772 \times 10^{-7}$, $1.83032585742 \times 10^{-7}$, $4.6074819334 \times 10^{-8}$
manufactured, asymmetric	$\max x_h - x^* $, 9/17/33 surfaces colloc_correct	$2.69073057043 \times 10^{-5}$, $7.20667053290 \times 10^{-6}$, $1.86266877703 \times 10^{-6}$
Solov'ev, 1–3 surfaces	zero of the collocated residual colloc_correct	exactly one, in boxes of radius 6.0×10^{-10} to 7.7×10^{-9}
rippled tokamak, $\iota = 1/2$ at $s = 0.625$	(2,1) harmonic of r_s at the converged state, 65/129/257/513/1025 surfaces, at 256 bits wdharm_b_correct ; over a box of relative width 10^{-14} , 65/129/257 surfaces dharm_correct	8.9208, 8.9310, 7.4567, 5.3557 and $3.8084 \times 10^{-5} \text{ T}^2$, each to a relative 2×10^{-12} ; over the box [8.81, 9.03], [8.51, 9.35], $[5.80, 9.11] \times 10^{-5} \text{ T}^2$
CTH-like, free boundary	largest jump of $\mu_0 p + B^2/2$ against $B_{\text{vac}}^2/2$ on NESTOR's grid, 25/385 surfaces check_bpcert_correct	$[7.556, 7.561] \times 10^{-4} \text{ T}^2$, $[4.9647, 4.9673] \times 10^{-4} \text{ T}^2$, for every vacuum pressure between NESTOR's and BIEST's
CTH-like, free boundary	the same jump over the whole boundary, 25/385 surfaces check_btcert_correct	at most $8.47 \times 10^{-3} \text{ T}^2$, at most $1.051 \times 10^{-2} \text{ T}^2$
CTH-like, free boundary	the largest jump on NESTOR's grid for every vacuum pressure within five times the two answers' disagreement, 25/49/97/193/385 surfaces check_bpcert_correct	at most 1.87, 1.89, 1.81, 1.73 and $1.66 \times 10^{-3} \text{ T}^2$
up_down_asym, $s = 0.125$, 17 surfaces	Mercier current term and DMerc merc_run_correct	$[-3.724, -3.716] \times 10^{-3}$ and $[-1.371, -1.205] \times 10^{-3}$, where the file gives -6.1×10^{-4} and $+1.8 \times 10^{-3}$ in floating point
up_down_asym, $s = 0.125$, 17 surfaces	the same with the file's centred radial differences merc_run_q_correct	$[-6.421, -6.366] \times 10^{-4}$ and $[1.711, 1.875] \times 10^{-3}$
Landreman's sheared equilibrium, 13/25/50/100/200 surfaces	largest component of $\mathbf{B} - \mathbf{B}^*$ at a 64×32 grid of a field period on the innermost, middle and outermost half-grid surfaces check_bpcert_correct	2.94×10^{-5} , 1.32×10^{-5} , 8.25×10^{-6} ; 1.36×10^{-5} , 2.93×10^{-6} , 2.00×10^{-6} ; 5.61×10^{-6} , 5.39×10^{-7} , 5.67×10^{-7} ; 4.35×10^{-6} , 1.12×10^{-7} , 1.02×10^{-7} ; 3.34×10^{-6} , 2.53×10^{-8} , $2.12 \times 10^{-8} \text{ T}$, where $ \mathbf{B}^* $ is 0.30 to 0.38 T
W7-X, $\beta = 4$ per cent, 49/97/193 surfaces	largest component of $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$ at a 64×64 grid of a field period on the surfaces $s = 1/4, 1/2, 3/4$ check_cert_correct ; the largest of its discrete harmonics at the 288 retained modes harm_encloses	at most 7.71, 7.73, $7.82 \times 10^{-2} \text{ T}^2$; at most 5.49, 5.49, $5.46 \times 10^{-4} \text{ T}^2$, where B^2 is about 7 T^2

Table 1: Certified results on the manufactured problems and on VMEC++ equilibria, with the soundness theorem behind each.